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Decomposition of the vehicle routing problem with time windows on the time dimension

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Abstract

The *Vehicle Routing Problem with Time Windows* (VRPTW) asks for the optimal set of routes to be performed by a fleet of vehicles to serve a set of customers within their assigned time windows. In this work, we propose a matheuristic for the VRPTW which utilizes the sub-problem constituted by optimizing only a selected time window of the VRPTW whereas the tours outside of this time window are regarded as fixed. We call this problem the *Single Time Window Vehicle Routing Problem* (STWVRP). For applying the STWVRP, we must assume that several customers are assigned to the same time window, i.e., the number of time windows is much smaller than the number of customers. An exact problem description of the STWVRP is given in the form of a Mixed-Integer Linear Programming formulation. We apply this exact formulation within a matheuristic for the VRPTW. The paper concludes with extensive computational experiments.

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1. Introduction

The *Vehicle Routing Problem with Time Windows* (VRPTW) asks for the optimal set of tours performed by a fleet of vehicles to serve a set of customers within their assigned time windows. This NP-hard problem, see [Lenstra and Kan \(1981\)](#), has become a vital tool in today's logistics and e-commerce landscape due to its wide range of applications. It constitutes an important variant of the classic Vehicle Routing Problem, which is among one of the most vivid fields in Operations Research, see [Wang and Wasil \(2021\)](#) for a recent analysis of the existing literature.

The VRPTW has a wide range of applications. It is, for example, the mathematical problem behind online grocery delivery services, referred to as *Attended Home Delivery* (AHD) problems in the scientific literature. The popularity of grocery delivery services surged during the recent Covid-19 pandemic, and their market share is projected to surpass

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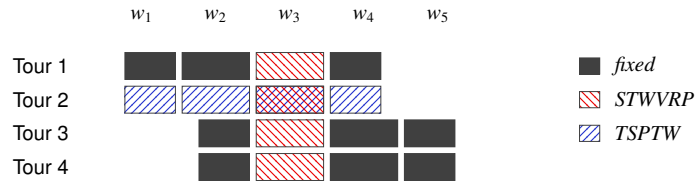


Figure 1: Comparison of the sub-problems of the VRPTW. Horizontal (*vehicle* dimension): TSPTW, vertical (*time* dimension): STWVRP. While TSPTW is concerned with a selected tour, the STWVRP is concerned with a selected time window. All other parts of the VRPTW are regarded as fixed.

20 % by 2025 (Redman (2020)). The main characteristic of AHD problems is that customers who ordered groceries online must be at home to receive their orders as perishable goods cannot be dropped off unattended like ordinary parcels. The company promises to deliver the orders. This requires arranging time windows during which customers are at home in order to receive a delivery. To this end, companies selling groceries online must deal with a VRPTW to guarantee a service that is satisfactory for their customers while being cost-effective for the company.

Further examples for vehicle routing problems with time window constraints are a) planning the installation of smart meters (Bucur et al. (2021)) or b) scheduling the visits of home-healthcare workers (Fikar and Hirsch (2017); Cissé et al. (2017)).

Another closely related category of optimization problems deals with dynamic variants of the VRPTW (see Pillac et al. (2013) for a review). These variants' dynamic nature stems from customer orders arriving when delivery vehicles are already traveling. These problem variants require periodic reoptimization (after the arrival of new orders). What we can observe is that the time windows of the VRPTW can serve as a natural partitioning of the planning horizon. While orders that are completed or are very close to being delivered cannot be rescheduled, orders assigned to upcoming time windows may still be rearranged (while ensuring that they stay within their time windows).

There is active research to develop exact algorithms for the VRPTW (see Kallehauge et al. (2007) for an overview). However, for large problem instances, as they occur in practice, (meta-)heuristics have to be used in order to derive approximate solutions in acceptable time.

Solving sub-problems of the VRPTW is also a common approach. In that sense, Savelsbergh (1992) introduces the *Traveling Salesperson Problem with Time Windows* (TSPTW) that addresses the optimal tour of a selected vehicle serving a corresponding sub-set of customers. While in Savelsbergh (1992) the TSPTW is solved heuristically, several strong *Mixed-Integer Linear Programming* (MILP) formulations are described, for example, in Ascheuer et al. (2001); Hungerländer and Truden (2018); Vu et al. (2020).

We observe that many of today's applications use only a limited set of time windows. For example, in AHD applications, customers pick a delivery time window from a set of alternatives that is provided by the supermarket chain. However, the proposed delivery time windows can still overlap. Providing a limited set of time windows has several benefits: a) It simplifies the planning process; b) the schedules are easier to read for a human dispatcher and the driver; c) in many applications (like AHD) arbitrary time windows can hardly be justified.

Typically, the VRPTW has two planning dimensions (besides the capacity constraints): a) the *vehicle* and b) *time*. Dividing the VRPTW with respect to the *vehicle* dimension creates a set of TSPTW instances where each one corresponds to one vehicle of the fleet. In this work, we explore the idea of dividing VRPTW instances with respect to the *time* dimension, i.e., cut them into "smaller" problems, where the delivery time windows specified by the company serve as naturally imposed "cut lines". To this end, we propose to solve the sub-problem constituted by optimizing only a selected time window of the VRPTW, whereas the vehicle tours outside of this time window are regarded as fixed. We call this problem the *Single Time Window Vehicle Routing Problem* (STWVRP). For applying the STWVRP, we must assume that several customers are assigned to the same time window (as already observed above), i.e., the number of time windows is much smaller than the number of customers. This assumption holds for most applications where customers select their delivery time window from a (limited) set of alternatives such as, for example, AHD. The STWVRP is only defined within the context of a feasible solution of a VRPTW, as it requires the arrival times at the first/last customer before/after the selected time window (of each tour) as input.

Finally, we illustrate the newly introduced STWVRP for a VRPTW that is related to four vehicles (tours) and five non-overlapping time windows in Figure 1.

2. Problem Description

We briefly introduce the required notation for defining the VRPTW. The set $\mathcal{W} = \{1, \dots, q\}$ contains all *time windows*, where each $w \in \mathcal{W}$ is defined through its begin and end time, $w = (B_w, E_w)$. We assume that the time windows are unique, in that there do not exist time windows $u, v \in \mathcal{W}$, $u \neq v$, with $B_u = B_v$ and $E_u = E_v$. Moreover, we are given a set of *customers* $\mathcal{O}$, $|\mathcal{O}| = p$, with a corresponding *order weight* function $c: \mathcal{O} \rightarrow \mathbb{R}^{>0}$, and a *service time* function $s: \mathcal{O} \rightarrow \mathbb{R}^{>0}$. Further, function $w: \mathcal{O} \rightarrow \mathcal{W}$ assigns a delivery time window to each customer during which a delivery vehicle must arrive. All vehicles are operated from the *depot* d , $\mathcal{V} := \mathcal{O} \cup \{d\}$. A *tour* $\mathcal{A} = (a_0, a_1, \dots, a_n, a_{n+1})$ contains customers $\{a_1, \dots, a_n\} \subseteq \mathcal{O}$, $|\{a_1, \dots, a_n\}| = n$, in the order in which they are visited and a_0 and a_{n+1} correspond to the depot d . Furthermore, each tour $\mathcal{A}$ has assigned *start* and *end times* that we denote as $start_{\mathcal{A}}$ and $end_{\mathcal{A}}$, respectively. Hence, the vehicle executing tour $\mathcal{A}$ can leave from the depot no earlier than $start_{\mathcal{A}}$ and must return to the depot no later than $end_{\mathcal{A}}$. The capacity of the vehicle is limited to $C_{\mathcal{A}} \in \mathbb{R}^{>0}$ weight units. The *travel times* are given by the function $t: \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}^{\geq 0}$, where we set the travel time from a customer $a \in \mathcal{O}$ or the depot d to itself to 0, i.e., $t(a, a) = 0$, $a \in \mathcal{V}$. Symmetry, i.e., $t(a, b) = t(b, a)$, $a, b \in \mathcal{V}$, and triangle inequalities, i.e., $t(a, c) \leq t(a, b) + t(b, c)$, $a, b, c \in \mathcal{V}$, are not guaranteed. A feasible VRPTW *schedule* $\mathcal{S} = \{\mathcal{A}, \mathcal{B}, \dots\}$ consists of $|\mathcal{S}| = m$ feasible tours, where each customer $a \in \mathcal{O}$ is included in exactly one tour.

Before defining the feasibility of a tour $\mathcal{A} \in \mathcal{S}$, we introduce the concept of earliest and latest arrival time e_{a_i} and ℓ_{a_i} . The earliest (latest) time at which the vehicle could arrive at customer a_i without violating time window and travel time constraints on the remaining tour $\mathcal{A}$ is given as follows (see, for example, Cwioro et al. (2019)):

$$\begin{aligned} e_{a_0} &:= start_{\mathcal{A}}, & e_{a_{j+1}} &:= \max \left\{ B_{w(a_{j+1})}, e_{a_j} + s(a_j) + t(a_j, a_{j+1}) \right\}, j \in [n-1]_0, & e_{a_{n+1}} &:= e_{a_n} + s(a_n) + t(a_n, a_{n+1}), \\ \ell_{a_{n+1}} &:= end_{\mathcal{A}}, & \ell_{a_{j-1}} &:= \min \left\{ E_{w(a_{j-1})}, \ell_{a_j} - t(a_{j-1}, a_j) - s(a_{j-1}) \right\}, j \in [n] \setminus \{1\}, & \ell_{a_0} &:= \ell_{a_1} - t(a_0, a_1). \end{aligned}$$

A tour $\mathcal{A}$ is *feasible* if both of the following conditions hold:

$$\begin{aligned} e_{a_i} &\leq E_{w(i)}, i \in [n] \quad \wedge \quad e_{n+1} \leq end_{\mathcal{A}}, & \text{TFEAS}(\mathcal{A}), \\ \sum_{i \in [n]} c(i) &\leq C_{\mathcal{A}}, & \text{CFEAS}(\mathcal{A}). \end{aligned}$$

While $\text{TFEAS}(\mathcal{A})$ ensures that the arrival times at each customer assigned to tour $\mathcal{A}$ are within their assigned time windows, $\text{CFEAS}(\mathcal{A})$ guarantees that the capacity of $\mathcal{A}$ is not exceeded.

The STWVRP is a sub-problem of the VRPTW, with the objective to minimizing the travel time of all tours during a selected time window $u \in \mathcal{W}$, of a schedule $\mathcal{S}$, and where the sub-tours comprised of the customers that are not included in u are regarded as being fixed. The problem is solely defined for feasible VRPTW schedules, i.e., it cannot be defined for a VRPTW instance for which no feasible solution is given.

We introduce some definitions for a VRPTW schedule $\mathcal{S}$. For each tour $\mathcal{A} \in \mathcal{S}$, $|\mathcal{A}| = n + 2$, we define the *first* and the *last* index of customers scheduled within a selected time window $u \in \mathcal{W}$ as $first(u, \mathcal{A}) := \min \{i \in [n]: \ell_{a_i} \geq B_u\}$ and $last(u, \mathcal{A}) := \max \{i \in [n]: e_{a_i} \leq E_u\}$, respectively. Both indices are not defined if there are no customers assigned to $u \in \mathcal{W}$. With that, each tour $\mathcal{A} \in \mathcal{S}$ can be defined by the following sub-tours (see Figure 2 for an illustration).

1. *Pre time window sub-tour*. Includes the start depot and all customers before the selected time window, i.e., $pre(u, \mathcal{A}) := \{a_i \in \mathcal{A}: i < first(u, \mathcal{A})\}$.
2. *Post time window sub-tour*. Includes all customers after the selected time window and the final depot, i.e., $post(u, \mathcal{A}) := \{a_i \in \mathcal{A}: i > last(u, \mathcal{A})\}$.
3. *Current time window sub-tour*. Contains the customers, which can be scheduled within the selected time window, i.e., $current(u, \mathcal{A}) := \{a_i \in \mathcal{A}: first(u, \mathcal{A}) \leq i \leq last(u, \mathcal{A})\}$. This is the part of the tour which is actually being altered when solving the STWVRP.

Note that the different sub-tours depend on the actual tour sequence and not just on the assignment of customers to time windows. Only in case of non-overlapping time windows the different sub-tours are solely dependent on the assignment of customers to time windows, e.g., $\text{current}(u, \mathcal{A}) = \{i \in \mathcal{A} \setminus \{a_0, a_{n+1}\} : w(i) = u\}$.

The STWVRP changes only the sub-tours that are defined by $\text{current}(u, \mathcal{A})$, $\mathcal{A} \in \mathcal{S}$. At the same time, the sub-tours defined by $\text{pre}(u, \mathcal{A})$ and $\text{post}(u, \mathcal{A})$ are not changed. Customers may be exchanged between $\text{current}(u, \mathcal{A})$ and $\text{current}(u, \mathcal{B})$ of different tours $\mathcal{A}, \mathcal{B} \in \mathcal{S}$, $\mathcal{A} \neq \mathcal{B}$ to minimize the travel times. In addition, the pre and post-time window sub-tours may be re-combined if the result is a feasible tour. Our model abstracts $\text{pre}(u, \mathcal{A})$ and $\text{post}(u, \mathcal{A})$ by creating “artificial” start and end depots which summarize the weights of their customers and define valid start and end times.

To formulate an MILP model of the STWVRP for a selected time window $u \in \mathcal{W}$, we first introduce further required sets, variables, and parameters:

- The set $O_u := \bigcup_{\mathcal{A} \in \mathcal{S}} \text{current}(u, \mathcal{A})$ denotes the set of customers scheduled within time window u .
- The sets D_s and D_f represent the m (artificial) start and end depots, respectively. The start (end) depot of tour $\mathcal{A} \in \mathcal{S}$ corresponds to the last (first) stop in $\text{pre}(u, \mathcal{A})$ ($\text{post}(u, \mathcal{A})$), i.e., $a_{\text{first}(u, \mathcal{A})-1}$ ($a_{\text{last}(u, \mathcal{A})+1}$). If u is the first (last) time window, then the start (end) depot of all tours is the actual start (end) depot.
- $\mathcal{F} := (\text{end}_1, \dots, \text{end}_r)$, $\text{end}_i < \text{end}_{i+1}$, $1 \leq i < r$, gives the sorted set of r unique end times, i.e., $\bigcup_{\mathcal{A} \in \mathcal{S}} \text{end}_{\mathcal{A}} = \bigcup_{1 \leq i \leq r} \text{end}_i$.
- The set $\mathcal{T} := \bigcup_{\mathcal{A} \in \mathcal{S}} (a_{\text{first}(u, \mathcal{A})-1}, a_{\text{last}(u, \mathcal{A})+1}) \subset D_s \times D_f$ gives all tuples of start and end depots, which are originally on the same tour. We split this set in r disjoint sets $T_1, \dots, T_r$ with $(s, f) \in T_i$, $1 \leq i \leq r$ if and only if $s \in D_s$ and $f \in D_f$ on a tour with the same end time end_i .
- The binary variables $x_{ij} \in \{0, 1\}$, $i \in \mathcal{P}_{\text{out}} := O_u \cup D_s$, $j \in \mathcal{P}_{\text{in}} := O_u \cup D_f$, $i \neq j$, with the interpretation: $x_{ij} = 1$ if customer j is visited directly after customer i , and 0 otherwise.
- The continuous variables y_i , $i \in O_u \cup D_s \cup D_f$, state the sum of the order weights of the tour from the start depot of the original tour up to customer or depot i .
- The continuous variables d_i , $i \in O_u \cup D_s \cup D_f$, give the arrival time at customer or depot i .
- The variables $q_i \in D_s$, $i \in O_u \cup D_s \cup D_f$, indicate the start depot of the tour containing customer or depot i .
- The parameters c_i , $i \in O_u$, where each c_i corresponds to the order weight $c(i)$ of customers i . Furthermore, the parameters c_i , $i \in D_s$, state the sum of the order weights of the customers of the pre time window sub-tour finishing at start depot i . Analogously, the parameters c_i , $i \in D_f$, give the sum of the order weights of the customers of the post time window sub-tour starting from end depot i .
- The non-negative parameters t_{ij} , $i \in \mathcal{P}_{\text{out}}$, $j \in \mathcal{P}_{\text{in}}$, $i \neq j$, give the service time $s(i)$ at customer i plus the travel time $t(i, j)$ from customer i to customer j .
- The parameter $C_{\max} = \max_{k \in [m]} C_k$ gives the maximum vehicle capacity.
- The parameters $M_1 = C_{\max} - \min_{i \in D_s} c_i + \max_{i \in \mathcal{P}_{\text{in}}} c_i$ and $M_2 = \max_{i \in D_f} \ell_i - \min_{i \in D_s} e_i + \max_{i \in \mathcal{P}_{\text{out}}, j \in \mathcal{P}_{\text{in}}, i \neq j} t_{ij}$ are used in the capacity

and arrival time constraints, respectively.

We can now formulate the STWVRP for time window u with $|\mathcal{F}| = 1$ as the following MILP:

$$\min \sum_{\substack{i \in \mathcal{P}_{\text{out}}, j \in \mathcal{P}_{\text{in}}, \\ i \neq j}} t_{ij} x_{ij} \quad (1)$$

$$\text{s.t.} \quad \sum_{\substack{i \in \mathcal{P}_{\text{out}}, \\ i \neq j}} x_{ij} = 1, \quad j \in \mathcal{P}_{\text{in}}, \quad (2)$$

$$\sum_{\substack{j \in \mathcal{P}_{\text{in}}, \\ j \neq i}} x_{ij} = 1, \quad i \in \mathcal{P}_{\text{out}}, \quad (3)$$

$$\sum_{\substack{i, j \in \mathcal{S}, \\ i \neq j}} x_{ij} \leq |\mathcal{S}| - 1, \quad \mathcal{S} \subset O_u, |\mathcal{S}| \geq 2, \quad (4)$$

$$y_j - y_i - M_1 x_{ij} \geq c_j - M_1, \quad i \in \mathcal{P}_{\text{out}}, j \in \mathcal{P}_{\text{in}}, i \neq j, \quad (5)$$

$$\min_{j \in \mathcal{D}_s} c_j + c_i \leq y_i \leq C_{\max} - c_i, \quad i \in \mathcal{O}_u, \quad (6)$$

$$y_i = c_i + C_{\max} - C_i, \quad i \in \mathcal{D}_s, \quad (7)$$

$$y_i = C_{\max}, \quad i \in \mathcal{D}_f, \quad (8)$$

$$d_j - d_i - M_2 x_{ij} \geq t_{ij} - M_2, \quad i \in \mathcal{P}_{\text{out}}, j \in \mathcal{P}_{\text{in}}, i \neq j, \quad (9)$$

$$\max(B_{w(i)}, \min_{j \in \mathcal{D}_s} e_j) \leq d_i \leq \min(E_{w(i)}, \max_{j \in \mathcal{D}_f} \ell_j), \quad i \in \mathcal{O}_u, \quad (10)$$

$$d_i = e_i, \quad i \in \mathcal{D}_s, \quad (11)$$

$$d_i = \ell_i, \quad i \in \mathcal{D}_f, \quad (12)$$

$$x_{ij} \in \{0, 1\}, \quad i \in \mathcal{P}_{\text{out}}, j \in \mathcal{P}_{\text{in}}, i \neq j, \quad (13)$$

$$y_i, d_i \in \mathbb{R}^+, \quad i \in \mathcal{O}_u \cup \mathcal{D}_s \cup \mathcal{D}_f. \quad (14)$$

The objective function (1) minimizes the sum of travel times during time window u . Expressions (2) guarantee that we visit all customers and the end depots exactly once, and expressions (3) ensure that we leave the start depots and all customers exactly once. Inequalities (4) are the well-known sub-tour elimination constraints. Note that preliminary computational experiments indicate that it is most efficient to use only constraints with $|S| \leq 3$. Anyway, sub-tours can not occur due to constraints (5) or constraints (9) that serve as the polynomially many Miller-Tucker-Zemlin sub-tour elimination constraints.

Inequalities (5) guarantee that each order weight is considered on its assigned tour. Further, inequalities (6) set the order weight variables y_i , $i \in \mathcal{O}_u$, as tight as possible. In expressions (7) we set y_i , $i \in \mathcal{D}_s$, to the sum of the weights of the orders of the pre time window sub-tour finishing at the start depot i plus the difference $C_{\max} - C_i$, where parameter C_i gives the capacity of the vehicle associated with start depot $i \in \mathcal{D}_s$. Expressions (8) set the capacity of the end depots to $C_{\max}$. In summary, (5) – (8) together with c_i , $i \in \mathcal{D}_f$, set to the sum of the order weights of the customers of the post time window sub-tour starting from end depot i , ensure that for each tour $\mathcal{A} \in \mathcal{S}$ holds CFEAS($\mathcal{A}$).

Inequalities (9) and (10) guarantee that the arrival times at customers consider all travel times correctly and are within the customers' assigned time windows. Expressions (11) set the arrival times d_i of our start depots $i \in \mathcal{D}_s$ to their earliest arrival times e_i . Expressions (12) ensure that the arrival times at the end depots $\mathcal{D}_f$ are early enough to serve the customers on the remainder of the tours within their assigned time windows. Note that we could also model constraints (8) and (12) as inequalities, but the use of expressions leads to a significant speed-up when solving the MILP. In summary, expressions (9) – (12) guarantee that for each tour $\mathcal{A} \in \mathcal{S}$ holds TFEAS($\mathcal{A}$).

In the above model ($|\mathcal{F}| = 1$) with expressions (1) – (14), all start depots are interchangeable, i.e., all combinations of pre and post time window sub-tours can occur. Now in case that $|\mathcal{F}| > 1$, we must avoid infeasible schedules, i.e., the occurrence of too long or too short tours. To prohibit such solutions, we adapt the capacity constraints by setting M_1 to $r \cdot C_{\max} + r - 1 - \min_{i \in \mathcal{D}_s} c_i + \max_{i \in \mathcal{P}_{\text{in}}} c_i$ and changing (6) – (8) as follows.

$$\min_{(s,f) \in T_1} c_s + c_i \leq y_i \leq r \cdot C_{\max} + r - 1, \quad i \in \mathcal{O}_u, \quad (6a)$$

$$y_i = c_i + t \cdot C_{\max} + t - 1 - C_i, \quad y_j = t \cdot C_{\max} + t - 1, \quad (i, j) \in T_t, 1 \leq t \leq r. \quad (7a, 8a)$$

It is ensured that no tour exceeds the capacity of its assigned vehicle and that customers $c \in \mathcal{O}_u$ can be re-assigned to the different sub-tours. However, due to inequalities (5), it is not allowed that $j \in \mathcal{D}_f$ is a successor of $i \in \mathcal{D}_f$ if $y_i > y_j$ and, hence, expressions (7a, 8a) guarantee that only start depots within the same T_t , $1 \leq t \leq r$, are arbitrary interchangeable. Hence, the resulting schedule is feasible. While the size of the VRPTW increases with the number of customers $|\mathcal{O}|$, the size of the STWVRP is only $|\mathcal{O}|/|W|$ (assuming that the customers are evenly distributed across the time windows and the time windows are non-overlapping). Therefore, the STWVRP can be applied to larger instances for which the VRPTW can not be solved exactly anymore.

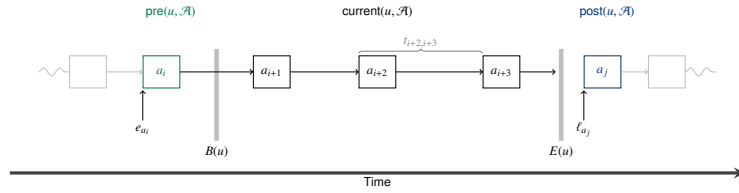


Figure 2: Illustration of the problem specific notation of the STWVRP. Showing one tour $\mathcal{A} \in \mathcal{S}$.

3. Computational Experiments

For our computational experiments, we apply a matheuristic for the VRPTW that utilizes the STWVRP MILP formulation. It starts with a heuristic solution of the considered VRPTW instance. Any feasible schedule for a given VRPTW instance can be the input of the matheuristic. The matheuristic selects a time window $u \in \mathcal{W}$ for which it calls the STWVRP, updates the overall VRPTW solution, and then repeats this procedure with another time window $v \in \mathcal{W}$, $u \neq v$. In this procedure, it is important in what sequence the time windows are selected. Finding the best sequence, therefore, is an important subject of our investigation. We compare three different sequences in which the STWVRP is applied to the different time windows $\mathcal{W}$. a) *one-time*: one iteration over all time windows, first to last time window, i.e., $w_1, w_2, \dots, w_q$; b) *two-times*: two iterations over all time windows, both times first to last time window, i.e., $w_1, w_2, \dots, w_q, w_1, w_2, \dots, w_q$; c) *random*: iterate over all time windows in random order (each time window once). We always warm-start the MILP for the currently considered time window with the sub-tours of the best schedule that has been found so far.

For evaluating the matheuristic, we use the benchmarking instances that were first presented by Cwioro et al. (2019). These instances contain 500 customers that are equally distributed among ten time windows. In this work, we consider two sets of time windows a) $\mathcal{W}_{\text{NO}}$: ten non-overlapping times windows of 60 minutes length, and b) $\mathcal{W}_{\text{OV}}$: ten time windows of 90 minutes length where a subsequent time window has a 30 min overlap with its predecessor. As input for the matheuristic, we use schedules that have been generated following the approach by Cwioro et al. (2019). The schedules are generated by sequential insertion of the customers into the schedule within their assigned time windows with a simple insertion heuristic (if possible), followed by (after each insertion) local search heuristics and solving the TSPTW exactly for each tour.

We implemented the matheuristic in Java, where we use Gurobi 8.1.0 as an IP-solver in single-thread mode. We run our experiments on an Ubuntu 14.04 machine equipped with an Intel Xeon E5-2630V3 @ 2.4 GHz 8 core processor and 132 GB RAM. We set time limits (one minute or 30 minutes) for each call of the STWVRP.

In Table 1, we show the results of the matheuristic comparing the three proposed improvement strategies for schedules where all vehicles have the same start and end times, i.e., we use constraints (6) – (8). Setting a time limit of one minute for each call of the MILP, the results show that for instances with 20 vehicles (tours), the improvement is much larger than for those with fewer vehicles (16 or 18). Contrary to our expectations, we observe faster runtimes of the matheuristic for large instances (more vehicles but a similar number of customers). Therefore, our approach can find better solutions in less time for those instances. Similar observations are made when setting a time limit of 30 minutes. Although the matheuristic starts from an optimized schedule, we observe improvements between 0.84 % and 15.7 %. Considering the AHD use case, already small improvements can make the difference between a profitable service and heavy losses. When comparing the results of the *one-time* and the *random* strategy, no major differences of the runtime and the improvements is noticeable. We conclude that the order in which the STWVRP is applied to the time windows can be neglected. Further, we observe that *two-times* strategy consistently yields better results than both other strategies where only one iteration across the time windows is done.

Next, we analyze the impact of using constraints (6a) (7a, 8a). Therefore, we use instances that differ from the ones used above only by the circumstance that vehicles have different start and end times ($|\mathcal{F}| > 1$). In the above scenario, the vehicle starts and ends 30 minutes before and after the first and the last time window. In this next analysis (using constraints (6a) (7a, 8a)), we shorten the vehicle drive times by 30 minutes and the vehicles leave the depot 30, 15, and 0 minutes prior to the start of the first time window. We report the corresponding computational results in Table 2. For the *one-time* and *random* strategy, improvements between 1.85 % and 11.57 % are reported. Again, the *two-times*

Table 1: Average results of five instances with the same vehicle start and end times of the matheuristic with the different improvement strategies. The first column p denotes the average customers considered within the instance and m defines the number of vehicles available. We report the improvement of the objective function (in percent) compared to the input schedule and the total required runtime.

p	m	$\mathcal{W}$	Time Limit [mm:ss]	one-time [%]	Runtime [hh:mm:ss]	two-times [%]	Runtime [hh:mm:ss]	random [%]	Runtime [hh:mm:ss]
450.8	16	$\mathcal{W}_{\text{NO}}$	01:00	2.32	00:10:38	3.51	00:21:28	2.05	00:10:38
491.6	18	$\mathcal{W}_{\text{NO}}$	01:00	7.47	00:09:31	9.50	00:18:26	6.95	00:09:40
498.0	20	$\mathcal{W}_{\text{NO}}$	01:00	11.59	00:08:02	15.52	00:13:22	11.53	00:07:01
450.8	16	$\mathcal{W}_{\text{NO}}$	30:00	4.41	04:43:13	6.07	09:11:59	3.68	04:54:58
491.6	18	$\mathcal{W}_{\text{NO}}$	30:00	9.10	03:54:45	11.80	06:43:58	9.42	03:25:26
498.0	20	$\mathcal{W}_{\text{NO}}$	30:00	13.35	01:42:16	15.70	02:23:11	12.37	01:46:45
459.6	16	$\mathcal{W}_{\text{OV}}$	01:00	0.84	00:11:14	1.24	00:22:25	1.04	00:11:10
495.8	18	$\mathcal{W}_{\text{OV}}$	01:00	2.60	00:11:11	4.83	00:22:18	3.35	00:11:07
499.2	20	$\mathcal{W}_{\text{OV}}$	01:00	4.52	00:11:20	7.60	00:22:34	5.87	00:11:14
459.6	16	$\mathcal{W}_{\text{OV}}$	30:00	2.97	05:00:57	4.02	10:01:58	2.61	05:01:01
495.8	18	$\mathcal{W}_{\text{OV}}$	30:00	6.93	04:56:32	9.77	09:53:51	7.18	05:01:09
499.2	20	$\mathcal{W}_{\text{OV}}$	30:00	9.47	04:54:08	12.21	09:16:46	11.10	04:56:12

Table 2: Average results of five instances with different vehicle start and end times of the matheuristic with the different improvement strategies. The first column p denotes the average customers considered within the instance and m defines the number of vehicles available. We report the improvement of the objective function (in percent) compared to the input schedule and the total required runtime.

p	m	$\mathcal{W}$	Time Limit [mm:ss]	one-time [%]	Runtime [hh:mm:ss]	two-times [%]	Runtime [hh:mm:ss]	random [%]	Runtime [hh:mm:ss]
449.0	16	$\mathcal{W}_{\text{NO}}$	01:00	2.36	00:10:38	3.15	00:21:12	2.36	00:10:29
485.6	18	$\mathcal{W}_{\text{NO}}$	01:00	5.90	00:10:04	8.41	00:19:37	5.77	00:10:13
492.0	20	$\mathcal{W}_{\text{NO}}$	01:00	9.27	00:09:41	12.98	00:18:12	9.39	00:09:04
449.0	16	$\mathcal{W}_{\text{NO}}$	30:00	3.90	04:56:43	5.15	09:46:19	4.05	04:50:42
485.6	18	$\mathcal{W}_{\text{NO}}$	30:00	9.12	04:15:10	11.83	08:02:05	8.23	04:37:02
492.0	20	$\mathcal{W}_{\text{NO}}$	30:00	11.57	03:35:26	14.59	06:11:21	10.60	03:30:55
459.8	16	$\mathcal{W}_{\text{OV}}$	01:00	1.85	00:11:00	2.54	00:22:02	1.76	00:10:59
494.4	18	$\mathcal{W}_{\text{OV}}$	01:00	2.65	00:11:11	3.62	00:22:23	2.75	00:11:10
496.6	20	$\mathcal{W}_{\text{OV}}$	01:00	3.61	00:11:15	5.18	00:22:31	3.89	00:11:14
459.8	16	$\mathcal{W}_{\text{OV}}$	30:00	3.62	05:01:02	4.72	10:02:02	3.88	05:00:59
494.4	18	$\mathcal{W}_{\text{OV}}$	30:00	6.10	05:01:17	9.87	09:59:05	6.76	05:01:14
496.6	20	$\mathcal{W}_{\text{OV}}$	30:00	7.46	04:57:02	10.27	09:55:29	8.28	04:57:18

strategy yields better results (2.54 % - 14.59 %), whereas the results of the one-time and the random strategy are more or less the same. For this setup, the number of vehicles does not seem to impact the runtimes.

In the above experiments, we notice that the two-times strategy with a time limit of one minute yields results that are close to the ones of the one-time strategy with a time limit of 30 minutes. For further investigations, we introduce the five-times strategy, i.e., five repetitions of the one-time strategy with a time limit of 30 seconds. In Table 3, we show the results of this five-times strategy applied to the same instances as for Table 1. We notice an improvement of the results compared to the two-times strategy with a time limit of one minute for both overlapping and non-overlapping time windows. Except one instance (non-overlapping time windows and 20 vehicles), the average improvement is less than the result of the two-times strategy with a time limit of 30 minutes.

Typically, (online) AHD setups allow rather little time to improve the schedules in between two incoming booking requests. Applying our MILP only to the time window that was last modified (by adding an order to it) can yield a continuous improvement of the schedule. Once the AHD system does not accept any new orders, the matheuristic can be applied (allowing larger time limits) to create the final delivery schedule.

4. Conclusion

We proposed a matheuristic for the *Vehicle Routing Problem with Time Windows* (VRPTW) which utilizes the sub-problem constituted by optimizing only a selected time window of the VRPTW whereas the tours outside of this time

Table 3: Average results of five instances with different vehicle start and end times of the matheuristic with the five-times improvement strategy. The first column p denotes the average customers considered within the instance and m defines the number of vehicles available. We report the improvement of the objective function (in percent) compared to the input schedule and the total required runtime.

p	m	$\mathcal{W}$	Time Limit [mm:ss]	five-times [%]	Runtime [mm:ss]
450.8	16	$\mathcal{W}_{\text{NO}}$	00:30	4.08	29:15
491.6	18	$\mathcal{W}_{\text{NO}}$	00:30	11.29	25:12
498.0	20	$\mathcal{W}_{\text{NO}}$	00:30	16.52	19:11
459.6	16	$\mathcal{W}_{\text{OV}}$	00:30	0.85	31:15
495.8	18	$\mathcal{W}_{\text{OV}}$	00:30	4.76	31:18
499.2	20	$\mathcal{W}_{\text{OV}}$	00:30	7.58	31:20

window are regarded as fixed. We call this problem the *Single Time Window Vehicle Routing Problem* (STWVRP). An exact problem description of the STWVRP by formulating a Mixed-Integer Linear Programming formulation is given and applied within the matheuristic. A concluding computational study illustrates that our approach is well-suited for different operational scenarios and is applicable in an online and an offline environment.

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